

NOVA

IMS

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Management
School

MDSAA

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Business Intelligence

Part 2

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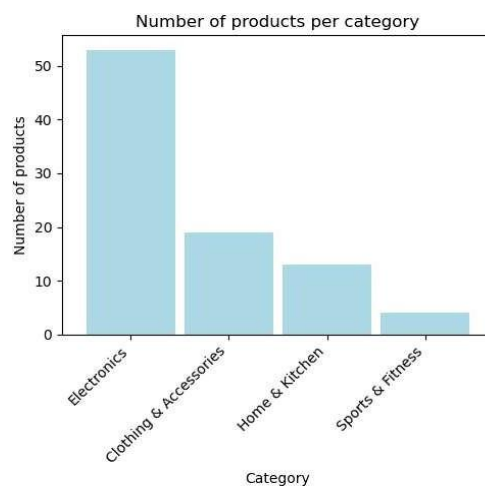
Figure 3.: Pipeline Design

1. PRESENTATION OF BUSINESS

G204All, a retail company founded in 2020, operates across Portugal as a business-to-customer (B2C) enterprise, offering a wide range of products directly to individual consumers through a direct sales model. The company manages 31 physical stores, 10 of which also provide an online shopping option, ensuring greater convenience and accessibility for its customers.

The products provided by the company span a diverse spectrum of categories, including Electronics, Clothing & Accessories, Home & Kitchen, and Sports & Fitness. G204All focuses on providing up-to-date, high-quality items that align with current trends and customer preferences, ensuring a wide selection to meet the varied needs of its consumers.

Figure 1: Number of Products per Category



Source: Authors

Once again, given the nature of a B2C business, the company collaborates with several point of supply (POS) partners which likely handle product sourcing or supply chain support.

G204All has a small yet diverse team of 15 operators, with roles ranging from Directors to Floor Staff. These team members are organized into different groups, each responsible for managing various aspects of the company's retail operations.

Due to its diverse product range, the company must place great emphasis on staying current with market trends. This includes keeping up with the latest technological advancements in the many sectors they are active or looking to be active in, helping them offer relevant and innovative products that meet their customers' changing needs. This approach helps G204All stay competitive and build trust with its customers.

2. BUSINESS PROBLEM

G204All is a growing retail company that operates both online and through physical stores. As the company expands, its administration faces challenges in obtaining timely and accurate insights to

support data-driven decision-making. The current reporting process is inefficient, lacks historical comparison capabilities, and does not allow for detailed segmentation of sales performance.

Currently, G204All's administration relies on CSV extracts from company servers, which refresh daily. However, the absence of a centralized data warehouse and proper data visualization tools limits the company's ability to analyze trends, compare performance across time periods, and optimize operations. One of the main issues G204All faces is the lack of visibility into daily sales data. The administration struggles to obtain a clear and real-time view of sales trends across different stores. Without a structured BI system, data remains scattered and underutilized, delaying strategic decision-making. Additionally, the company lacks an efficient way to track sales performance over time, making it difficult to analyze sales per month, quarter, and year. This prevents the identification of seasonal trends and business growth patterns, while also making year-over-year sales comparisons more challenging.

Another critical challenge is understanding product performance. G204All cannot easily determine which products drive the most revenue and which underperform. The absence of structured reporting on product categories and subcategories results in inefficiencies in inventory management and marketing strategies. Similarly, evaluating store performance across multiple locations remains difficult, as there is no centralized view to compare sales performance and identify underperforming stores or regions.

G204All also struggles with gaining deeper customer insights. The company lacks the ability to build detailed customer profiles, which limits the potential for targeted marketing and personalized promotions. Without an analysis of purchasing behaviors, the company is missing opportunities for upselling and cross-selling.

Finally, there are security concerns regarding data access. G204All requires Row-Level Security (RLS) to ensure that sales data is only accessible to authorized users based on their store location or management level. The current data handling process does not guarantee proper security, potentially exposing sensitive sales figures to unauthorized employees.

To address these challenges, G204All needs a structured BI solution supported by a dimensional model (star schema). By implementing a Data Warehouse (DWH) and Power BI dashboards, the company will be able to gain real-time insights into sales performance, identify high-performing products and stores, improve inventory and marketing strategies, and ensure secure, role-based access to data. This initiative will provide G204All with the data-driven decision-making capabilities needed to enhance operational efficiency and drive business growth.

3. BUSINESS QUESTIONS AND DISCOVERY PROCESS

To enhance G204All's analytical capabilities and support data-driven decision-making, the following business questions have been identified. These questions focus on understanding sales performance, customer behavior, and operational efficiency. Each of them is linked to specific measures that will be used to derive insights and improve strategic planning.

The table 1 below outlines the key business questions along with the corresponding metrics for analysis.

Table 1: Business Questions

Business Question	Measures
What are the sales per month, quarter, and year?	Gross Sales
How does the current year's performance compare to the previous year in terms of profits?	Sales growth
What are the top N clients in terms of total sales?	Sales amount (gross sales)
What are the top N best-performing stores in terms of quantity of products sold?	Sales amount and quantity
Which are the N most sold products?	Sales quantity (volume)
What is the average quantity of products ordered and how much each customer spent?	Sales amount and quantity
What are the top operators when it comes to the number of products sold and total gross sales?	Sales amount and quantity
How do sales differ between online and physical stores, and which channel generates more revenue and profit?	Sales amount and profit margin
Which suppliers contribute the most to total sales?	Sales amount
Which product categories and subcategories generate the highest sales volume and revenue?	Sales amount and quantity
What are the busiest hours of the day for sales?	Sales volume

Source: Authors

4. SOURCE DATA

The data used has been retrieved by the company and provided through seven .csv files: Locations, Operators, Points of Supply, Products, Sales (Retail), Stores A and Stores B.

Locations: This file contains various municipalities in Portugal, each associated with a unique identifier. The file gives a hint of possible interest of the company in these locations.

Operators: Includes information about the company's workforce. A unique identifier, their first and last name, their email, gender and most importantly, their role in the company as well as the team they operate in.

Point of Supply: Regards, as mentioned before, partners who are likely to handle product sourcing or supply chain support. Contains an unique identifier as well as a name and an email.

Products: The file contains a unique identifier, SUK, for each product, as well as its name, category and subcategory.

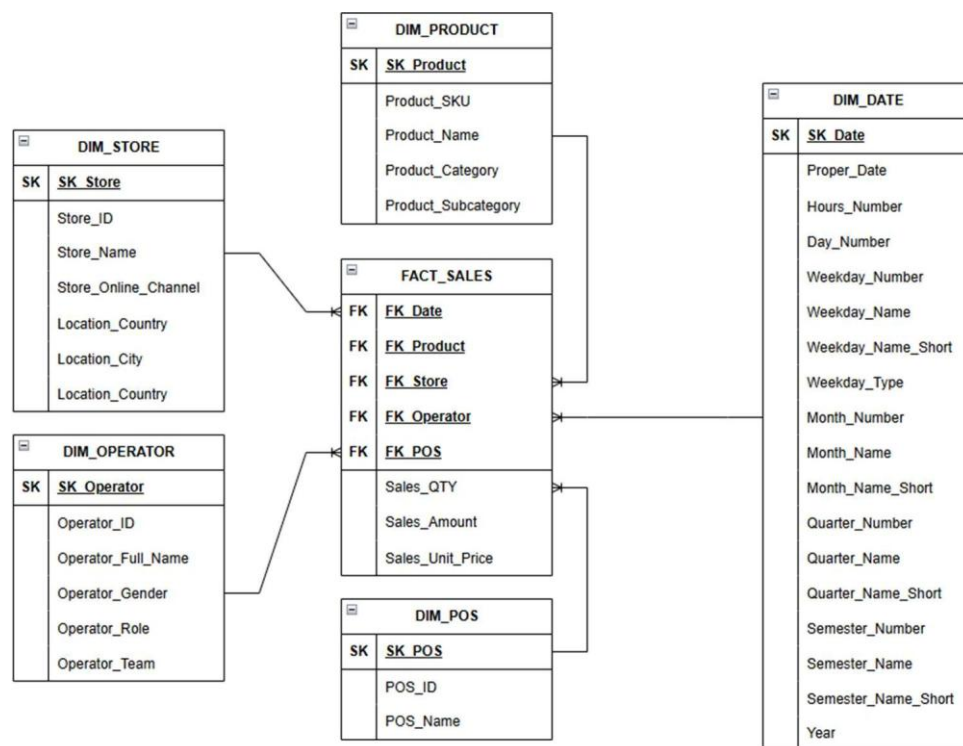
Sales (Retail): Detailed record of all sales made by each product, including the datetime, the product identifier and its quantity, the store, point of supply, location, transacted currency, amount and the unique identifier of the operator.

Stores A and Stores B: These two files have similar information and could be joined into just one. They indicate a unique identifier for each store, their name, location and if they have an online presence. Stores A has information from stores 1 to 20 and Stores B has information about the remaining.

5. DATA MODELING

5.1. STAR SCHEMA

Figure 2: Start Schema



Source: Authors

5.2. DIMENSIONAL MODEL EXPLANATION

Table 2: DIM_DATE Table

Column Name	Data Type	Description
SK_Date	INT	Surrogate Key
Proper_Date	DATE	01/01/2025
Hours_Number	INT	18
Day_Number	INT	1
Weekday_Number	INT	1
Weekday_Name	VARCHAR	Sunday
Weekday_Name_Short	VARCHAR	Sun
Weekday_Type	VARCHAR	Weekend or Weekday
IS_Special	BOOLEAN	Holiday
Month_Number	INT	1
Month_Name	VARCHAR	January
Month_Name_Short	VARCHAR	Jan
Quarter_Number	INT	1
Quarter_Name	VARCHAR	1º Quarter
Quarter_Name_Short	VARCHAR	1Q
Semester_Number	INT	01
Semester_Name	VARCHAR	1º Semester
Semester_Name_Short	VARCHAR	1S
Year	INT	2025

Source: Authors

Table 3: DIM_PRODUCT Table

Column Name	Data Type	Description
SK_Product	INT	Surrogate Key
Product_SKU	VARCHAR	Original identifier of the product in the database
Product_Name	VARCHAR	Product Name
Product_Category	VARCHAR	Product Category
Product_Subcategory	VARCHAR	Product Subcategory

Source: Authors

Table 4: DIM_STORE Table

Column Name	Data Type	Description
SK_Store	INT	Surrogate Key
FK_Store_Location	INT	Store's Location
Store_ID	INT	Unique Identifier
Store_Name	VARCHAR	Store Name
Store_Online_Channel	BOOLEAN	Yes or No

Source: Authors

Table 5: DIM_OPERATOR Table

Column Name	Data Type	Description
SK_Operator	INT	Surrogate Key
Operator_ID	INT	Unique Identifier
Operator_Full_Name	VARCHAR	Operator's Full name

Operator_Gender	VARCHAR	Male or Female
Operator_Role	VARCHAR	Operator's Role
Operator_Team	VARCHAR	Operator's Team

Source: Authors

Table 6: DIM_POINT OF SUPPLY Table

Column Name	Data Type	Description
SK_POS	INT	Surrogate Key
POS_ID	INT	Unique Identifier
POS_Name	VARCHAR	Point of supply's name

Source: Authors

Table 7: FACT_SALES Table

Column Name	Data Type	Description
SK_Sales	INT	Surrogate Key
FK_Date	INT	Foreign Key
FK_Product	INT	Foreign Key
FK_Store	INT	Foreign Key
FK_Operator	INT	Foreign Key
FK_Location	INT	Foreign Key
FK_POS	INT	Foreign Key
Sales_QTY	VARCHAR	Number of products bought
Sales_Amount	INT	Monetary value of the transaction

Sales_Unit_Price

FLOAT

Monetary unit of each value

Source: Authors

After having all the tables defined, the following hierarchies were established:

- Store Dimensional Table: Location_County > Location_City > Location_Country
- Product Dimensional Table: Product_Name > Product_Category > Product_Subcategory
- DateTime Dimensional Table: Hours > Day > Month > Quarter > Semester > Year

6. DATA MODELLING METHODOLOGY

To build an efficient and structured data warehouse that supports insightful analysis and data-driven decision-making, G204All has adopted the Kimball methodology for dimensional modeling. This approach ensures that data is organized in a way that optimizes reporting and analytical capabilities while maintaining high performance and ease of use.

The first step in this methodology is to identify the business process that the data warehouse will support. In the case of G204All, the primary process being analyzed is the retail sales process, which follows a B2C (business-to-consumer) model. This process includes customer transactions at different store locations, both physical and online, involving various products, payment methods, and operators.

Once the business process is defined, the next step is determining the granularity of the fact table, which refers to the level of detail at which data is stored. For G204All, the data warehouse will store sales transactions at the individual sales level, capturing details such as sales per hour, store, location, product, and operator. This level of granularity allows for high-detail analysis, such as hourly, daily, weekly, monthly, semesterly and yearly trends, and supports comparisons across different business segments.

With the granularity established, the next step is identifying the dimensions that provide context for the facts and allow for flexible and intuitive data analysis. The final step in the Kimball methodology is defining the fact table, which contains numerical values that quantify the business process.

Additional measures, such as discounts applied, profit margins, and tax amounts, can also be included depending on the company's analytical needs.

By following the Kimball methodology, G204All's data warehouse will be structured for efficient querying, reporting, and business intelligence analysis. The star schema design, with a well-defined fact table and multiple dimensions, ensures that decision-makers can easily access and analyze critical business insights. With granular data storage and structured dimensions, G204All will be able to track sales trends, monitor store and employee performance, and make informed strategic decisions that drive business growth.

7. MAIN COMPONENTS

The developed dimensional model follows the star schema design, which optimizes performance for analytical queries and reporting. It consists of one fact table (FACT_SALES) and multiple dimension tables, each providing context for the sales transactions.

The DIM_DATE dimension captures attributes like proper date, hours, day, weekday, month, quarter, semester and year, allowing time-based analysis of sales trends and performance. Weekdays are also included in order to have an indicator for special dates such as holidays.

The DIM_PRODUCT dimension defines all products sold by the company and classifies them into categories and subcategories. It contains attributes such as surrogate key, SKU (business key), product name, category and subcategory, enabling detailed product performance tracking.

The DIM_STORE dimension represents different store entities, differentiating between physical and online stores. It includes attributes such as store ID, store name, location reference and an online channel indicator, facilitating store-level performance comparisons.

The DIM_OPERATOR dimension tracks the employee's handling transactions, allowing performance analysis at the operator level. It includes operator ID, name, email, gender, role, and team information, providing insights into employee contributions and efficiency.

The DIM_LOCATION dimension identifies store locations, including county, city and country. This enables regional sales analysis and supports decisions related to store expansion and market penetration.

The DIM_POS (Point of Supply) dimension captures supply chain details and links sales with supplier information. It includes attributes like a unique identifier, supplier name and contact email, supporting supply chain analysis and optimization.

At the core of the data warehouse is the FACT_SALES table, which stores quantitative measures reflecting the business's sales activities. The table's surrogate key, SK_Sales, serves as a unique identifier combining all foreign keys from the dimension relationships. This fact table links all dimensions through these foreign keys and enables detailed analysis and reporting. Key facts include Sales_QTY (for analyzing product demand patterns), Sales_Amount (for revenue calculation and trend analysis), and Sales_Unit_Price (for pricing strategy evaluation), supporting comprehensive financial tracking.

The relationships in the model ensure data connectivity, with the FACT_SALES table as the central point linking all dimensions. This structure enables efficient query performance and supports various business intelligence insights, such as time-based sales trends, product performance evaluation, store comparisons, employee performance tracking, geographical insights, and supply chain analysis.

8. THE ETL PROCESS

The ETL process, built in Microsoft Fabric, automated the extraction, cleansing, and loading of source data into a star schema. By leveraging a shared workspace, the team collaboratively transformed unprocessed data into analyzable dimensions, enabling data-driven decision-making and

actionable insights.

Lakehouse

The first step of the ETL process was to create a Lakehouse to upload all the csv files needed regarding the Retail4All business process. From 'Get Data' were uploaded all the six files, Locations.csv, Operators.csv, Point_of_Supply.csv, Products.csv, Sales (Retail).csv, StoresA.csv, and lastly StoresB.csv.

Warehouse

A dimensional Data Warehouse named WH_Retail_Group20 was designed and implemented. This warehouse was developed to support advanced analytics and reporting on sales operations, enabling structured, high-performance querying and integration with visualization tools like Power BI.

The main goal of the data warehouse is to centralize sales data across different dimensions, including date, product, store, operator, and point of supply, following the star schema model already mentioned. This design improves query efficiency and provides a scalable architecture for future data expansion.

The data warehouse consists of one central fact table (FACT_SALES) and multiple dimension tables (DIM_POINT_OF_SUPPLY, DIM_OPERATOR, DIM_STORE, DIM_PRODUCT, DIM_DATE) each capturing a specific perspective of the sales activity.

To ensure a clean environment, all existing tables were dropped prior to the creation of the schema using SQL DROP TABLE IF EXISTS. The tables were then created with proper naming conventions, surrogate keys (SKs) as primary keys, and foreign key references to ensure data integrity and relational consistency.

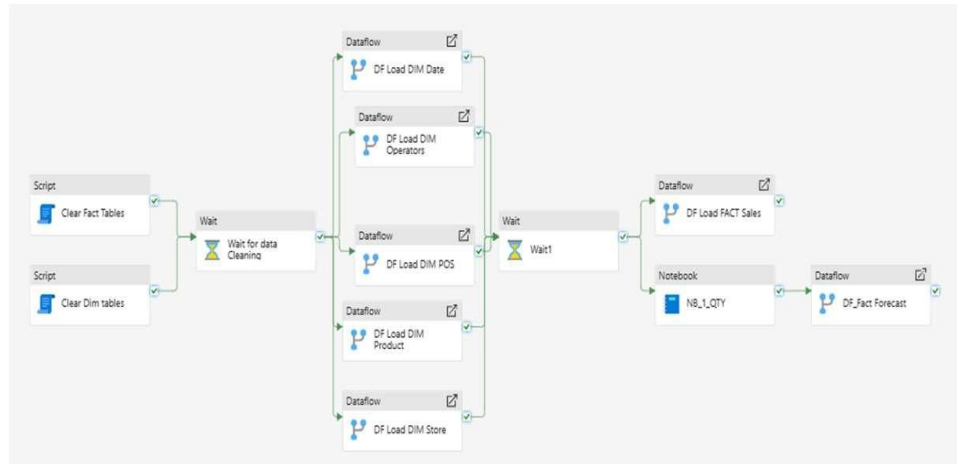
Dataflows

A Dataflow was created for each dimension table, named as: 'DF_Date', 'DF_Operators', 'DF_POS', 'DF_Product', 'DF_Store', and 'DF_Sales'. This implementation represented the initial phase of the Transformation stage in our ETL pipeline. Each Dataflow sourced its raw data from the corresponding tables in our Lakehouse, which was established in the previous phase. We applied customized transformation logic to each Dataflow based on its specific dimensional requirements, including data cleansing, key generation, and attribute standardization. The detailed transformation steps for each dimension will be elaborated in subsequent sections of this report.

Pipeline

The final step to implement is the load process. To achieve this, we started with the following steps. First, we implemented a full load. This begins by creating two scripts. For both scripts, the data warehouse is selected in the settings. In the script for the dimensions, we're deleting the existing dimensions, and in the script for the fact table, we're deleting the data from the fact table. This is done to prevent overwriting or multiplying data when the pipeline is run. Next, because these processes can take different amounts of time to complete, we're inserting a wait step. This wait ensures that both the dimension and fact table deletion processes finish before moving on. After this wait, the dataflows for the existing dimensions are added. Once again, following these dataflows, wait steps are connected. This is to ensure that all dimension dataflows finish running before the next step in the pipeline. Then the DF load Fact Sales was connected to the wait and the notebook with the forecast too.

Figure 3: Pipeline Design



Source: Authors

Notebooks

Two Python notebooks were created in Microsoft Fabric ("NB_1_Group20" and "NB_2_Group20") to predict sales using ARIMA and SARIMAX models. A small data preparation process was made, aligning datetime with quantity values and filling missing quantities with zeros. Finally, the models predict the quantity sales for the last 60 days.

9. DATA ENGINEERING

Although most data transformation steps are specific to each dimension, some are general, such as renaming the columns to fit the appropriate names in accordance with the Star Schema, most dataflows also needed to push the first row to headers to rightfully display the column names. The specific transformation steps were:

In DF_Load DIM Date, the fact table was imported, and all columns but 'Datetime' were deleted, an index 'SK_Date' was created: Then, 'Datetime' column was split into 'Proper_Date' and 'Time'. Through the 'Proper_Date' column, there were several columns created, including 'Day_Number', 'WeekDay_Number', 'WeekDay_Name', 'WeekDay_Name_Short', 'WeekDay_Type', 'Month_Number', 'Month_Name', 'Month_Name_Short', 'Quarter_Number', 'Quarter_Name', 'Quarter_Name_Short', 'Semester_Number', 'Semester_Name', 'Semester_Name_Short' and 'Year'. Through the 'Time' column, the 'Hours' column was created.

In 'DF_Load DIM Operators', an 'SK_Operators' index was created, a custom column 'full_name' was created through merging the columns 'first_name' and 'last_name' (with a space in between them).

In 'DF_Load DIM POS' and 'DF_Load DIM Product' there were also indexes columns created called 'SK_Point_Of_Supply' and 'SK_Product' respectively, being the only changes made.

'DF_Load DIM Store' encompasses three original tables, 'Stores A', 'Stores B' and 'Locations', the

first two were appended to create a new table 'Append' containing information of all stores present in the database. This new table suffered transformation in the 'Online' column where answers were passed to binary responses, 'Y' and 'S' to 1 and 'N' to 0. The missing values were filled and assumed as not having an online store. Finally, the 'Append' table was merged with the 'Locations' on the 'Locations_ID' variable through a left join, creating the 'Store' table.

In 'DF_Load FACT Sales', an index was created 'FK_Sales' and the table was merged with the 'Products' table through a left join to include the 'FK_Products' variable, needed in the fact table.

10. REFERENCES

Star Schema - Kimball, R., & Ross, M. (2013). *The data warehouse toolkit: The definitive guide to dimensional modeling* (3rd ed.). Wiley.

Dimensional Model - Kimball, R., & Ross, M. (2013). *The data warehouse toolkit: The definitive guide to dimensional modeling* (3rd ed.). Wiley.